DHC-Net: A Remote Sensing Object Detection Under Haze and Class Imbalance

Yuanyuan Li, Qiying Ling, Yiyao An, Hongpeng Yin, Xinbo Gao, Zhiqin Zhu, and Peng Han

***Abstract*— Object detection in remote sensing images is crucial in numerous fields; however, it becomes highly challenging under adverse weather circumstances. Given that previous remote sensing image object detection methods were designed based on normal weather conditions and ideal datasets, they are not beneficial for detection under real-world haze conditions and with class-imbalanced data. In this work, an adaptive dehazing centroid contrastive network (DHC-Net) is proposed to address the aforementioned issues. This network consists of an adaptive dehazing module and a centroid-guided contrastive learning approach. The adaptive dehazing module learns the image content to generate adaptive dehazing parameters, thus alleviating the influence of haze on the quality of remote sensing images. The centroid-guided contrastive learning approach is particularly designed to address the issue of imbalanced datasets. Integrating centroid vectors with actual samples in each training batch guarantees that each class is sampled at least once, effectively preventing the undersampling of minority classes. Moreover, dynamic weighted sampling based on prediction confidence guides the model to give priority to smaller classes, remarkably improving its ability to handle imbalanced data. Extensive experiments on the DOTA-v2.0, DOTA-v2.0Haze, RTTS, and HazeNet datasets demonstrate that DHC-Net is outstanding in handling haze conditions in remote sensing data, substantially enhancing target detection accuracy, even in the presence of imbalanced object classes. The source code will be available at https://github.com/Linghuaqian1/DHC_Net**

***Index Terms*— Contrastive learning, deep learning, dehazing, object detection, remote sensing image.**

I. Introduction

With the rapid advancements in Earth observation, satellite remote sensing data has become a crucial and strategic national resource. Remote sensing image object detection is essential for various aspects of the national economy, social development, and national security. It is widely used in agriculture [1], land-cover classification [11], environmental monitoring [2], urban planning [3], [4], disaster detection and response [5], national defense [6], resource exploration [7], land-cover change detection [8], [9], [10], and so on. Most remote sensing object detection methods are typically developed under the assumption of ideal conditions with ample and balanced samples [16], [17]. However, this assumption often does not apply to real-world scenarios, where remote sensing data frequently suffers from class imbalance and complex climatic conditions [18]. Fig. 1 provides examples of object detection in hazy conditions with class imbalance. As illustrated, when the image is enhanced to address adverse weather effects and the focus is shifted toward minority classes, more hidden details about previously obscured and misclassified objects can be uncovered. To address these issues, it is important to examine the specific challenges posed by haze interference and class imbalance and to explore strategies to overcome them.

1) In complex climatic haze scenarios, remote sensing images often face various disturbances such as atmospheric scattering and changes in lighting. These issues result in degraded image quality, which can manifest as color distortions, reduced contrast, and loss of fine details in small objects [14], [15]. These quality problems directly affect the accuracy and reliability of object detection algorithms, making it difficult to effectively identify and locate objects in real-world conditions. As a result, haze interference poses a significant challenge that limits the practical use of remote sensing object detection technology.
2) Deep learning models depend heavily on extensive and expensive data annotation. The supervised learning approach is particularly sensitive to data imbalance [19], often leading to a focus on accurately classifying majority classes while ignoring minority classes. This results in much lower recognition performance for minority classes. Given the diversity of geographic regions and varying land-cover types, class imbalance in remote sensing datasets can significantly reduce the overall performance of supervised models [20]. Thus, class imbalance remains a major challenge impacting the accuracy of remote sensing object detection methods, especially for small sample sizes in real-world conditions.

To tackle haze issues in complex climatic conditions, some studies use image dehazing as a preprocessing step [12]. Dehazing techniques not only improve image quality and

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Fig. 1. Under hazy and class-imbalanced conditions, our method is capable of producing clearer images with sharper edges around object boundaries, resulting in higher-confidence detection outcomes while reducing the number of missed instances. The blue circles indicate missed detections. (a) Haze image. (b) Baseline. (c) DHC-Net.

visual clarity but are also vital for systems that need to function well in various weather conditions [13], [14]. Recently, learnable image preprocessing methods have become a better option [15]. Unlike traditional methods that use manually designed filters or transformations, learnable methods apply deep learning to automatically determine the best preprocessing steps from the data itself. However, even with advanced dehazing models, improved restoration does not always translate to better object detection performance [16]. Additionally, adding dehazing to detection processes might slow down detection speed because of the extra task involved.

To tackle class imbalance, some studies use contrastive learning-based object detection methods [21], [22]. These approaches leverage large amounts of unlabeled data to learn similarities between objects and use sample relationships to create more effective training strategies [23], thereby improving overall detection performance in remote sensing images [24], [25]. However, these contrastive learning methods mainly focus on unsupervised pretraining or clustering, which can struggle with distinguishing subtle category differences due to the lack of labeled data. Additionally, most of these methods perform contrastive learning at the image level, treating all other images as negative samples and images with random augmentations as positive samples [26]. By using a memory bank, they introduce a large number of negative samples, which helps in achieving better feature representation [27]. Yet, when dealing with class-imbalanced datasets, these methods may face challenges: majority class samples might lack enough negative samples, while minority class samples might not have sufficient positive samples, leading to reduced model performance on small samples.

This study addresses the dual challenges of haze interference and class imbalance in remote sensing images, where existing research is often confined to single-issue solutions and lacks supporting adaptable datasets. In summary, this article introduces a dehazing centroid contrastive network (DHC-Net), a new method for remote sensing image object detection designed for complex climatic conditions. The DHC-Net combines an adaptive dehazing module with centroid-guided contrastive learning to tackle both haze-related image quality issues and class imbalance problems. The adaptive dehazing module uses a small convolutional neural network to predict dehazing parameters, generating clearer images for detection and minimizing the negative effects of low-quality images on model performance. The centroid-guided contrastive learning method calculates the centroids of each category in every training batch, ensuring that each category is represented at least once and providing a sufficient balance of positive and negative samples. This effectively addresses the issue of small-sample undersampling caused by data imbalance. Furthermore, the method employs a dynamic weighting strategy to focus on challenging areas in the predicted images, thereby dynamically balancing the class imbalance in the image. Through these approaches, it adeptly resolves the issues of low-quality images under haze conditions and class imbalance in remote sensing datasets. The main contributions of this article are as follows.

1) This article introduces the DHC-Net, an object detection network that combines adaptive dehazing with centroid-guided contrastive learning to detect objects in remote sensing images affected by haze. Additionally, a synthetic dataset named DOTA-v2.0Haze is introduced, which retains the class imbalance characteristics of DOTA-v2.0 and includes remote sensing images obscured by haze, to validate the effectiveness of the proposed method.
2) To address the impact of haze and fog on object detection in remote sensing images under complex weather conditions, an adaptive dehazing module is introduced to achieve adaptive dehazing of remote sensing images.
3) To address the issue of class imbalance, a centroid-guided contrastive learning module is proposed. This module achieves improved accuracy of model pairs by balancing the sampling probability of a few classes using the proposed center of mass calculation method.

II. Related Work

A. Remote Sensing Object Detection

Remote sensing object detection uses automated image recognition to identify and locate ground objects, with applications in urban planning, environmental monitoring, military reconnaissance, and disaster response. Current object detection models in remote sensing generally improve performance through three main strategies: rotating anchor boxes and generating candidate boxes, optimizing label assignment and sampling strategies, and refining loss function regression.

Methods that use rotating anchor boxes and candidate box generation effectively model the rotation angles of objects, creating rotated candidate boxes that align with the target object's orientation. This approach enhances detection accuracy across various directions and angles. Xie et al. [40] proposed the dual-focused detector (DFDet) to tackle challenges related to contextual knowledge and angle regression sensitivity in aerial images. Xie et al. [41] also introduced the oriented region proposal network (Oriented RPN) for generating high-quality oriented proposals efficiently. Additionally, Xie et al. [42] developed the object activation network (OAN), which boosts detection efficiency and accuracy by reducing the number of patches. However, despite these improvements in detection performance, these methods often struggle with high-density small targets and complex backgrounds, leading to issues like

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positive–negative sample imbalance and increased computational costs.

Methods that focus on label assignment and sampling strategies improve detection performance by optimizing the distribution of positive and negative samples as well as the selection of samples during training. Zhao and Li [50] proposed orientation-adaptive sampling learning (OASL) to address consistency issues when detecting objects at arbitrary orientations. Hou et al. [59] introduced shape-adaptive selection (SAS) and shape-adaptive measurement (SAM), where SAS emphasizes object shape information related to aspect ratios and calculates optimal Intersection-over-Union (IoU) thresholds for objects with varying aspect ratios. Although these methods enhance detection performance to some degree, the backgrounds in remote sensing images are often complex and variable. This is especially true in scenarios with multiscale and densely distributed targets, where the label assignment strategies may struggle to accurately distinguish between targets and backgrounds, leading to unreasonable sample allocation and negatively affecting detection accuracy.

Methods based on loss functions directly impact the optimization direction and detection accuracy of models by designing more effective loss functions during training. Yu et al. [52] developed H2RBox-v2 to bridge the gap between HBox-supervised detection and RBox-supervised detection. Wang et al. [54] proposed a multitask pretraining (MTP) paradigm for comprehensive pretraining on remote sensing datasets. Yang et al. [60] introduced a method for addressing the angles of rotated boxes through classification, successfully overcoming issues related to angular periodicity and edge exchange. However, this approach requires longer bit lengths for angle encoding, complicating the prediction layer and increasing the model's parameter count and computational overhead [60]. To address this issue, Qian et al. [61] designed focal rotated IoU (FRIoU) to measure the overlap between two rotated boxes. This method calculates the offsets of the top and right vertices of the rotated box from the midpoints of the top and right edges of its minimum enclosing rectangle, incorporating these offsets into Focal-EIoU to enhance the precision of the rotated boxes [61]. While these methods optimize the regression accuracy of rotated boxes, there is still room for improvement in handling angular periodicity and detecting small targets, and the complexity of loss calculations can lead to computational bottlenecks.

In summary, while these methods have made significant advancements in angle regression, proposal generation, and loss functions, they still struggle to address the diversity and complexity of conditions in remote sensing scenarios. This highlights the need for future research to enhance the robustness and adaptability of methods while effectively managing imbalanced data and extreme weather conditions. In contrast to general remote sensing object detection methods, the DHC-Net considers the specific characteristics of foggy climates and the imbalanced nature of remote sensing image samples.

B. Object Detection in Hazy Weather Conditions

In complex climatic conditions, such as fog and haze, the performance of object detection networks is often significantly challenged. To address this issue, existing research methods can be primarily categorized into three main types: two-stage methods, single-stage methods, and domain adaptation methods.

Two-stage methods usually involve a stepwise optimization process for image preprocessing and object detection [55], [56], [58], [68]. Liu et al. [62] proposed a deep neural network solution that cascades two modules for image denoising and various advanced tasks, updating only the denoising network through joint loss via backpropagation. For example, Guo et al. [12] introduced zero-reference deep curve estimation (Zero-DCE), which enhances images by estimating specific curves using deep networks. Although this approach has theoretical advantages, it often does not achieve a proportional improvement in detection accuracy in practical applications.

Single-stage methods perform feature extraction and object detection simultaneously in a unified framework. The AOD-Net is one of the pioneers that integrates image dehazing with object detection [63]. Qin et al. [13] introduced the feature fusion attention network (FFA-Net), which employs channel and pixel attention mechanisms for dehazing. Song et al. [14] developed DhazeFormer, an image-dehazing model based on the Swin Transformer architecture. Liu et al. [15] proposed an adaptive image enhancement approach tailored to each image to improve detection performance. Kalwar et al. [16] introduced the gated differentiable image processing (GDIP) module, a domain-agnostic network architecture. Feng et al. [67] proposed HazeNet, a large-scale dataset specifically designed for drone-based object detection in hazy scenes, and developed a depth-conditioned detector (DeCoDet) to integrate prior knowledge of object scale and clarity under varying depths and haze conditions, thereby enhancing detection performance. Wang et al. [69] introduced an effective detection paradigm named TogetherNet, which combines image restoration and object detection through dynamic enhancement learning to improve the performance of advanced detectors in adverse weather conditions. While these methods for enhancing image quality have made progress, improvements in image quality do not always lead to better detection performance.

Domain adaptation methods aim to enhance the model's robustness under adverse weather conditions by narrowing the gap between the source domain and the target domain. Hnewa and Radha [28] proposed a multiscale domain adaptation approach that leverages the domain transformation relationships among images captured under varying conditions to achieve feature invariance across scales and domains. Although this method performs well on conventional images, the increased computational costs in remote sensing image detection in complex environments may impact detection speed and accuracy.

In summary, while current image enhancement and domain adaptation methods have made some strides in complex climatic conditions, their performance gains in practical remote sensing image detection tasks are still limited. These methods often come with higher computational costs, making it difficult to improve accuracy without slowing down detection speed.

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C. Contrastive Learning for Object Detection

In recent years, unsupervised contrastive learning has gained increasing attention for its use in pretraining for image classification and object detection tasks. Currently, object detection models typically improve performance through two main approaches: feature contrastive learning and cross-domain, multitask contrastive learning.

Feature contrastive learning methods improve object detection performance by comparing local or global features. These methods often utilize images from different perspectives or transformations, or they focus on specific local regions in an image to extract and enhance relevant feature information. This approach helps in strengthening the representation of useful features, ultimately boosting detection accuracy. Liu et al. [22] introduced a self-supervised dense contrastive learning method utilizing the Swin Transformer, employing parallel global and dense projection heads to capture both global and local features. Yuan et al. [64] tackled the challenge of handling small-sized instances, particularly the low overlap between priors and object regions and the lack of discriminative information, by proposing a two-stage framework based on coarse-to-fine pipelines and feature imitation learning. Xie et al. [65] addressed the poor performance of current unsupervised pretraining methods in object detection and image classification by introducing DetCo, a self-supervised object detection method that enhances instance-level dense prediction tasks through multilevel supervision of intermediate representations and contrastive learning between global images and local patches. However, feature contrastive learning can sometimes overlook global contextual information when extracting local features, resulting in insufficient recognition capability in complex scenes.

Cross-domain and multitask contrastive learning methods focus on comparing different data domains or tasks to enhance the model's generalization ability. Zhang et al. [21] proposed an object detection method that utilizes contrastive learning representations, which improved the performance of remote sensing object detection. Lv et al. [23] developed a self-supervised contrastive learning approach that leverages sample relationships to create more effective training strategies, thereby enhancing remote sensing object detection performance. Wu et al. [66] addressed the significant decline in object detection performance under adverse weather conditions, such as fog, by proposing a robust object detection algorithm based on contrastive learning. This algorithm consists of a contrastive learning module and an object bounding box prediction module, learning representation vectors by maximizing the consistency between different augmented views of the same foggy image and using these representations to predict the bounding boxes of each object. However, cross-domain and multitask contrastive learning can encounter challenges in domain adaptation due to distribution differences between the source and target domains, and multitask optimization may lead to interference among tasks.

In summary, while these contrastive learning methods have advanced object detection, they mainly focus on exploring contextual information in unsupervised conditions, often overlooking the category relationships that exist in supervised settings. Additionally, these approaches generally involve high computational costs and tend to perform poorly on imbalanced datasets, which makes it difficult to effectively tackle issues related to insufficient negative or positive samples in small-class instances. As a result, the model's performance in detecting small-class targets remains limited.

III. Dataset

The effectiveness of deep learning-based object detection and dehazing methods relies significantly on the quality and scale of available datasets. In practical applications, remote sensing increasingly requires data that mirrors complex weather conditions, especially haze, to ensure reliable target detection and analysis under challenging circumstances. However, the lack of a dedicated dataset tailored to these conditions hinders advancements in remote sensing tasks under adverse atmospheric conditions. Constructing such a dataset is especially difficult due to the high costs and logistical barriers to collecting large-scale, high-quality data across varying weather conditions. The absence of standardized benchmarks further complicates remote object detection in hazy environments. Currently, dehazing datasets primarily include real hazy images captured with specialized equipment or synthetic datasets generated by adding haze to clear-weather images [29], [30], [31], [32]. While real data more accurately captures the intricacies of weather effects, its acquisition cost is high. Conversely, synthetic haze, despite some limitations, effectively replicates natural hazy conditions at a lower cost, making it a viable alternative for simulating real-world scenarios [33], [34], [35]. Collecting and annotating extensive remote sensing data in adverse weather is resource-intensive, further complicating the development of large-scale real hazy remote sensing datasets.

To tackle this issue, and inspired by synthetic dehazing datasets, this article introduces the DOTAv-2.0Haze dataset, based on the DOTAv-2.0 dataset, to allow researchers to more effectively evaluate and improve detection model performance in hazy conditions. DOTA-v2.0, released by Wuhan University, is the third dataset in the DOTA-v2.0 series, comprising 11,268 remote sensing images collected from Google Earth, the GF-2 satellite, and other aerial satellites. It contains a total of 1 793 658 instances across 18 categories. The specific formula for haze synthesis is outlined as follows:

$$I(x) = J(x)e^{-\beta}d(x) + A(1 - e^{-\beta}d(x)) \tag{1}$$

$$d(x) = -0.04 * \rho + \sqrt{\max(\text{row}, \text{col})} \tag{2}$$

where ρ represents the Euclidean distance from the current pixel to the image center, row and col denote the number of rows and columns in the image, respectively, $A = 0.05$, $\beta = 0.01*i+0.05$, and i is a random number set between 0 and 9 to achieve varying degrees of haze synthesis in the image. This process results in randomly dehazed images, as illustrated in Fig. 2 with the original and synthesized images. To simulate the variations found in real weather conditions, this dataset uses a hybrid approach that combines hazy and clear images in a 2:1 ratio. This method helps the model learn to capture

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Fig. 2. Comparison of haze synthesis effects. (a) DOTA-v2.0. (b) DOTA-v2.0Haze.

TABLE I
DETAILED INFORMATION ON THE DOTA-V2.0HAZE DATASET

class	instances	class	instances
PL	23930	LV	89353
SH	251883	SV	1235658
ST	79497	HC	893
BD	3834	RA	6809
TC	9396	SBF	2404
BC	3556	SP	20095
GTF	4933	CC	3887
HB	29581	AP	5905
BR	21433	HP	611
Training	268627	Test/Tes-dev	353346
Validation	81048	Test-challenge	1690637

features without being overly influenced by haze. DOTAv-2.0Haze is the first large-scale dataset created for remote sensing image detection in hazy environments. Table I presents detailed information on the distribution of the DOTA-v2.0Haze dataset. Fig. 3(a) illustrates the distribution of each category in DOTA-v2.0Haze, revealing a significant imbalance in samples, with the number of Helipad instances differing by a factor of 410 compared to small vehicles. In summary, DOTAv-2.0Haze possesses the following characteristics: 1) it establishes the first large-scale dataset for detecting remote sensing images in complex haze environments; 2) it integrates both real and synthetic data, offering a more accurate simulation of hazy scenarios; and 3) it effectively addresses the challenges associated with collecting and annotating data under complex weather conditions.

IV. PROPOSED METHOD

A. Overview

The overall framework of the proposed DHC-Net is illustrated in Fig. 4. This network utilizes CSPDarkNet [53] as its backbone. Additionally, an adaptive dehazing module, which does not impact detection speed, is incorporated in the preprocessing section. Furthermore, the model enhances its focus on small-sample categories through centroid-guided contrastive learning on feature maps and real images sampled from the backbone network.

The following summary provides a detailed overview of the adaptive dehazing module and the centroid-guided contrastive learning method. The adaptive dehazing module employs a small neural network, called the PK-Model, to predict input parameters, thereby improving image quality and detection performance. Compared to traditional contrastive learning methods, centroid-guided contrastive learning has significant advantages in feature accuracy, interimage contrast, and feature utilization efficiency. This approach allows the model to extract features more reliably and consistently. By integrating these advanced methods, DHC-Net effectively manages haze interference in complex weather conditions and addresses class imbalance issues. Experimental results show that DHC-Net not only performs exceptionally well on standard datasets but also demonstrates strong and adaptive capabilities in challenging environments, leading to improved detection accuracy and speed.

B. Adaptive Dehazing Module

In this article, the traditional atmospheric scattering model is used to generate haze images. The specific formula of the atmospheric scattering model is as follows:

$$I(x) = J(x)t(x) + A(1 - t(x)) \tag{3}$$

where $I(x)$ is the haze image, $J(x)$ is the original scene image, A represents the global atmospheric light, and $t(x)$ is the transmission matrix.

The transmission matrix $t(x)$ is calculated as follows:

$$t(x) = e^{-\beta d(x)} \tag{4}$$

where β is the atmospheric scattering coefficient and $d(x)$ is the distance of the shot. Thus, the computation of the dehazed image is obtained as follows:

$$J(x) = \frac{1}{t(x)} I(x) - A\frac{1}{t(x)} + A. \tag{5}$$

From (1), it can be observed that to recover the ideal clean image, two key parameters need to be determined: the global atmospheric light A and the transmission matrix $t(x)$. To reduce potential errors, estimating the global atmospheric light A and the transmission matrix $t(x)$ separately might

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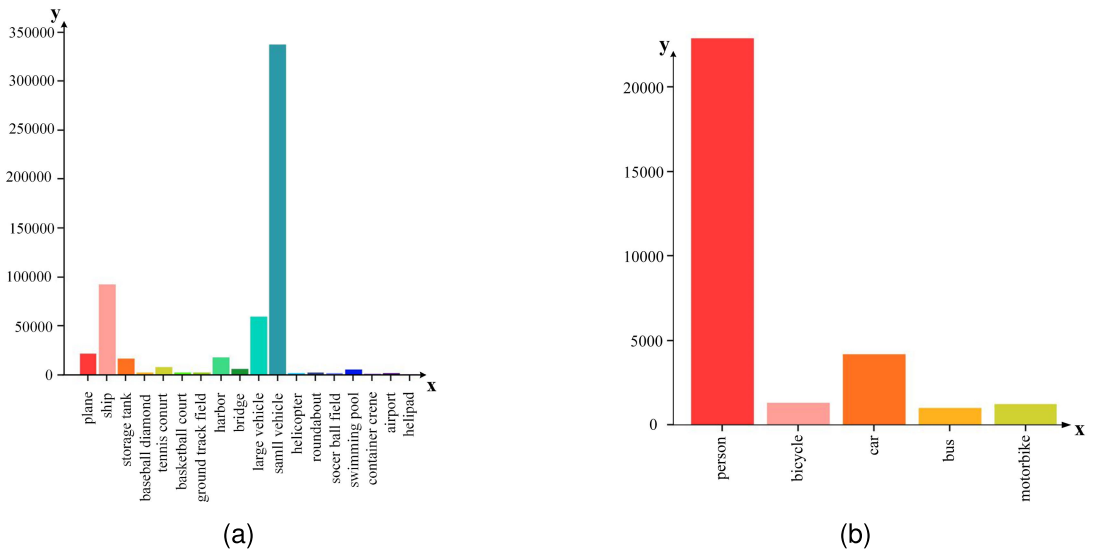


Fig. 3. Distribution of dataset categories. (a) DOTA-v2.0Haze. (b) RTTS.

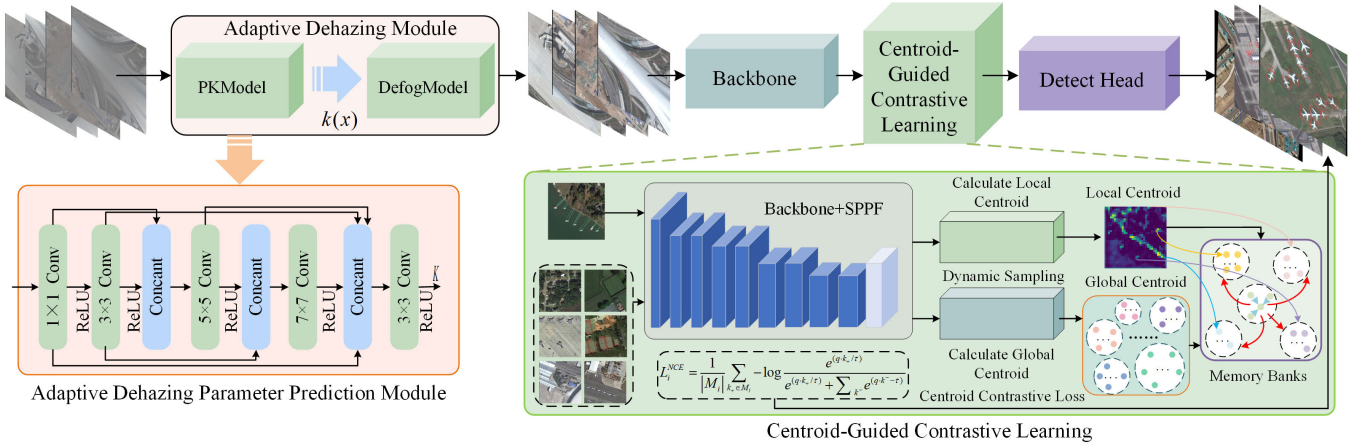


Fig. 4. Overall architecture of the DHC-Net. The adaptive dehazing module comprises the parameter prediction module, PKModel, and the dehazing module. The detailed structure of the PKModel is shown in the upper left corner. Centroid-guided contrastive learning aims to utilize label information to uncover latent interimage relationships and intraimage instance relationships to obtain richer category information. The implementation details are depicted in the upper right corner.

lead to increased inaccuracies. Therefore, both parameters are jointly computed using the following formula:

$$J(x) = K(x)I(x) - K(x) + b \tag{6}$$

$$K(x) = \frac{\frac{1}{t(x)}(I(x) - A) + (A - b)}{I(x) - 1} \tag{7}$$

where $K(x)$ represents the combined variable of the global atmospheric light A and the transmission matrix $t(x)$ and b is a preset constant bias. From (5), it can be seen that $K(x)$ can be calculated based on the haze image $I(x)$.

Therefore, an adaptive parameter prediction network can be constructed, where the predicted parameters adjust with the input haze image. Substituting (9) into the calculation yields the dehazed image. By substituting (9) $k(x)$, the dehazed image is computed.

Traditional dehazing model parameters are often fixed for image dehazing, lacking consideration for varying degrees of haze conditions. Therefore, to comprehensively address varying degrees of haze conditions in different scenes, a small, lightweight CNN network named the PKModel is proposed to predict and estimate dehazing model parameters. The PKModel, as depicted on the left side of Fig. 4, consists of five convolutional layers of varying sizes. These layers generate feature maps at different scales, which are then fused using three connection layers to integrate deep and shallow features. This enables the network to fully capture image information, thereby predicting dehazing model parameters $K(x)$ as accurately as possible.

C. Centroid-Guided Contrastive Learning

Remote sensing image object detection relies heavily on large volumes of annotated data. Therefore, leveraging the advantages of massive annotated datasets in remote sensing, this study explores the application of contrastive learning in remote sensing image detection under complex weather conditions in a fully supervised manner. Using supervised

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information from the data to form global class centroids helps mitigate the problem of undersampling minority class samples. Employing a weighted approach for dynamically sampling predicted images directs more attention to challenging instances within images that are difficult to recognize. Using centroids of the same class as positive samples and centroids of other classes as negative samples for contrastive learning with predicted images further explores potential relationships between images and instance relationships within them, enhancing the richness of category information. Finally, by utilizing a contrastive loss function to increase interclass distances and reduce intraclass variances, the model's feature representation capability is enhanced. The specific implementation process is detailed in Algorithm 1.

Specifically, to reduce computational load and align with the diverse characteristics of remote sensing objects, the centroid of the class in a remote sensing image is defined as the average value of all pixels within a single image that belongs to this class. Assuming the feature map size is $F \in \mathbb{R}^{C\times H\times W}$, C, H, W representing the dimensions, height, and width of the feature map, respectively, the calculation process for the i type of the regional center is as follows:

$$\mathrm{BoxAv}g_c(i) = \frac{\sum_{y=y^i_{\min}}^{y^i_{\max}} \sum_{x=x^i_{\min}}^{x^i_{\max}} \mathrm{imageP}(x,y)}{\left(x^i_{\max} - x^i_{\min} + 1\right) \times \left(y^i_{\max} - y^i_{\min} + 1\right)} \tag{8}$$

$$R_i = \frac{\sum_{i=1}^{N} \mathrm{BoxAv}g_c(i)}{N} \tag{9}$$

where i represents the i class, $\mathrm{imageP}(x, y)$ denotes the pixel value at position (x, y) on the feature map, N indicates the number of predicted bounding boxes for the class i, $(x^i_{\min}, y^i_{\min})$ represents the top-left coordinates of each prediction box, and $(x^i_{\max}, y^i_{\max})$ represents the bottom-right coordinates of each prediction box.

From the analysis of the above formula, it is evident that when the object detection network processes small samples, the predicted categories may contain more errors, leading to centroids constructed with incorrect feature information. These errors can cause the centroid construction process to deviate from the true category. To address this issue and ensure the network focuses on small sample categories, a dynamic sampling method weighted by confidence is proposed to construct the class centroid. In addition to feature maps and prediction results, this method also incorporates real label information to extract the global centroid. To avoid erroneous information contamination, a simple weight allocation using prediction confidence is employed. Ultimately, the centroid of the class i is defined as

$$R_i = \frac{\mathrm{BoxAv}g_c(i)((1 - c(x,y)) \cdot EP_i + HP_i)}{\sum_{y=y^i_{\min g}}^{y^i_{\max}} \sum_{x=x^i_{\min}}^{x^i_{\max}} \mathrm{imageP}(G_{\mathrm{Box}}(x,y) = i)} \tag{10}$$

where the real image is denoted as $G \in \mathbb{R}^{H\times W}$, the feature map as $F \in \mathbb{R}^{C\times H\times W}$, and the prediction map as $P \in \mathbb{R}^{C\times H\times W}$. C represents the feature dimension, N is the number of classes, and $c_{i,j} \in [0, 1]$ is the confidence score of the instance class. The weight for positive samples is $(1 - c_{i,j})$, while the weight for true samples is 1. EP_i represents simple positive samples and HP_i represents difficult positive samples.

After constructing the centroids for all classes for each image using (12), a comparison with the global centroid of the class i is necessary. The contrastive learning loss is specifically defined as follows:

$$L_i^{\mathrm{NCE}} = \frac{1}{|M_i|} \sum_{k_+\in M_i} -\log \frac{e^{(q\cdot k_+/\tau)}}{e^{(q\cdot k_+/\tau)} + \sum k^- e^{(q\cdot k^- - \tau)}} \tag{11}$$

where M_i represents the memory bank, which collects the centroids of the class i from the entire training set. k_+ denotes the positive samples, while k^- represents the negative samples from other classes in the memory bank.

During training, the overall loss function for the object detection algorithm is optimized to improve the accuracy of the prediction boxes. The cross-entropy loss function for classification and the DFL loss to enhance generalization are retained. Additionally, the aforementioned contrastive loss is incorporated. The final loss function for object detection can be expressed as

$$L = L_{\mathrm{box}} + L_{\mathrm{cls}} + L_{\mathrm{dfl}} + \lambda L_{\mathrm{NCE}} \tag{12}$$

where λ is the weight of the contrastive loss.

Algorithm 1 Compute Contrastive Loss Function

Input: Feature map F, Ground truth image G, Predicted image P
Output: Contrastive loss function L

```
Initialize memory bank M
for each ground truth image G_i in the training batch do
    for each class c do
        Sample pixels belonging to class c from G_i
        Compute the centroid BoxAvg_c(i) for class c
        Compute the average R_i for class c in the batch
        where imageP(x, y) denotes the pixel value at position (x, y) in image G_i
        Update the centroid for class c in the memory bank M
    end for
end for
for each predicted image P_i do
    Perform weighted sampling based on prediction confidence
    for each class c do
        Compute the centroid R_i for class c
        where P_{p,c} denotes all pixels in image P_i belonging to class c
    end for
end for
Use the global centroid of the same class from memory bank M as positive samples and different classes as negative samples to compute the contrastive loss L
return L
```

V. Experiments

A. Experimental Setup

1) Datasets: To more comprehensively evaluate the effectiveness of the proposed model, the following two datasets were selected: RTTS and HazeNet.

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TABLE II
DETAILED INFORMATION ON THE VOC TRAINING AND TESTING DATASETS AND THE RTTS DATASET

Dataset	image	person	bicycle	car	bus	motorcycle	total
V_n_tv	8111	13256	1064	3267	822	1052	19561
V_n_ts	2734	4528	337	1201	213	325	6604
RTTS	4322	7950	534	18413	1838	862	29577

TABLE III
DETAILED DESCRIPTION OF THE HAZENET DATASET

Split	Images	Car	Truck	Bus	All
Train	8000	242195	11626	10730	264511
Val	1000	31562	1508	1490	34560
Test	2000	60026	2553	2743	65322
RDDTS	600	18220	489	587	19296

1) *RTTS:* The RTTS dataset consists of 4,322 natural haze images across five categories: person, car, bus, bicycle, and motorcycle. The RTTS dataset exhibits significant class imbalance, with the number of car instances far surpassing that of other categories, while instances of bicycles and motorcycles are extremely scarce, as illustrated in Fig. 3(b). Additionally, to further evaluate the method's generalizability, the VOC_haze dataset is created by extracting images of persons, cars, buses, bicycles, and motorcycles from VOC2007 and VOC2012. The training dataset is named VOC_norm_trainval (V_n_tv), and the testing dataset, which includes the five categories, is called VOC_norm_test (V_n_ts). Detailed information about V_n_tv, V_n_ts, and the RTTS datasets is provided in Table II.
2) *HazeNet [67]:* The HazeNet dataset is a large-scale dataset specifically designed for object detection in haze scenarios using drones. It contains 110 600 images and 383 000 real-world instances, distributed across three object categories: car, truck, and bus. These instances were collected from natural haze environments and normal scenes, with synthetic haze effects applied to simulate adverse weather conditions. Table III presents detailed information about the distribution of the HazeNet dataset, where RDDTS represents real haze images.

2) Model Training and Evaluation Metrics: The model in this article is constructed based on PyTorch and trained on a GeForce RTX4090 GPU with a batch size of 4. The backbone network, pretrained on the COCO dataset, is used for transfer learning on the RTTS and DOTA-v2.0Haze datasets. During training, data augmentation techniques such as Mosaic, CopyPaste, RandomPerspective, MixUp, and RandomFlip are employed. The SGD optimizer with weight decay is used, with a momentum of 0.937 and an initial learning rate of 0.01, over 50 epochs. To evaluate the detection performance of the algorithm model, several commonly used evaluation metrics for object detection are adopted: precision, recall, mAP50, and FPS.

TABLE IV
COMPARISON EXPERIMENTS ON THE DOTA-V2.0HAZE DATASET

Method	Precision(%)	Recall(%)	mAP50(%)
YOLOv8 [53]	70.8	55.3	57.3
DAYolo [28]	66.8	46.7	49.2
IA-YOLO [15]	69.3	46.6	50.5
GDIP [16]	70.2	48.2	52.1
Our	72.8	56.0	62. 0

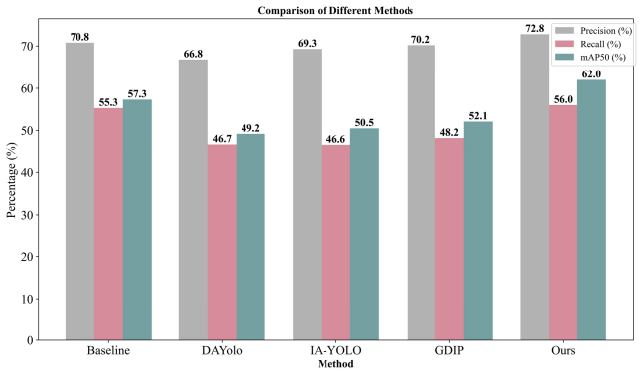


Fig. 5. Comparison of test performance of each method on DOTA-v2.0Haze.

B. Comparison With SOTAs

This section presents a qualitative and quantitative analysis of the experimental results of the proposed method across four different datasets, highlighting its feasibility and effectiveness. The detailed experimental setup and results are as follows.

1) Experimental Result on the DOTA-v2.0Haze Dataset: In this study, YOLOv8, renowned for its excellent detection speed and accuracy, is selected as the baseline method. To demonstrate the effectiveness of our approach, we conducted a comprehensive comparison of current representative methods on the DOTA-v2.0Haze dataset. DAYOLO [28] and IA-YOLO [15] address haze detection under adverse weather conditions from different perspectives; the former focuses on the distribution discrepancy between source and target data, while the latter achieves adaptive enhancement through a differentiable image processing module. Both methods provide valuable references for our approach. Table IV presents the results of our method alongside several mainstream approaches on the DOTA-v2.0Haze test set. The table indicates that our proposed method outperforms others in terms of accuracy, recall, and mAP50, showcasing a significant performance advantage. Specifically, compared to [53], DAYOLO [28], IA-YOLO [15], and GDIP [16], our method achieves an accuracy of 72.8%, with a recall increase of 7.8% points and a mAP50 improvement of 9.9% points. Notably, our method's performance in mAP50 far exceeds that of the other methods. This clearly demonstrates that our approach excels in the accuracy and overall performance of remote sensing image detection under complex haze conditions. To reflect the advantages and disadvantages of different methods in each performance index and to facilitate the rapid assessment and comparison of the performance of each method, the experimental results of each method were plotted as bar charts. The specifics of each method's indicators are shown in the bar chart in Fig. 5. The visual analysis of Fig. 5 shows that the

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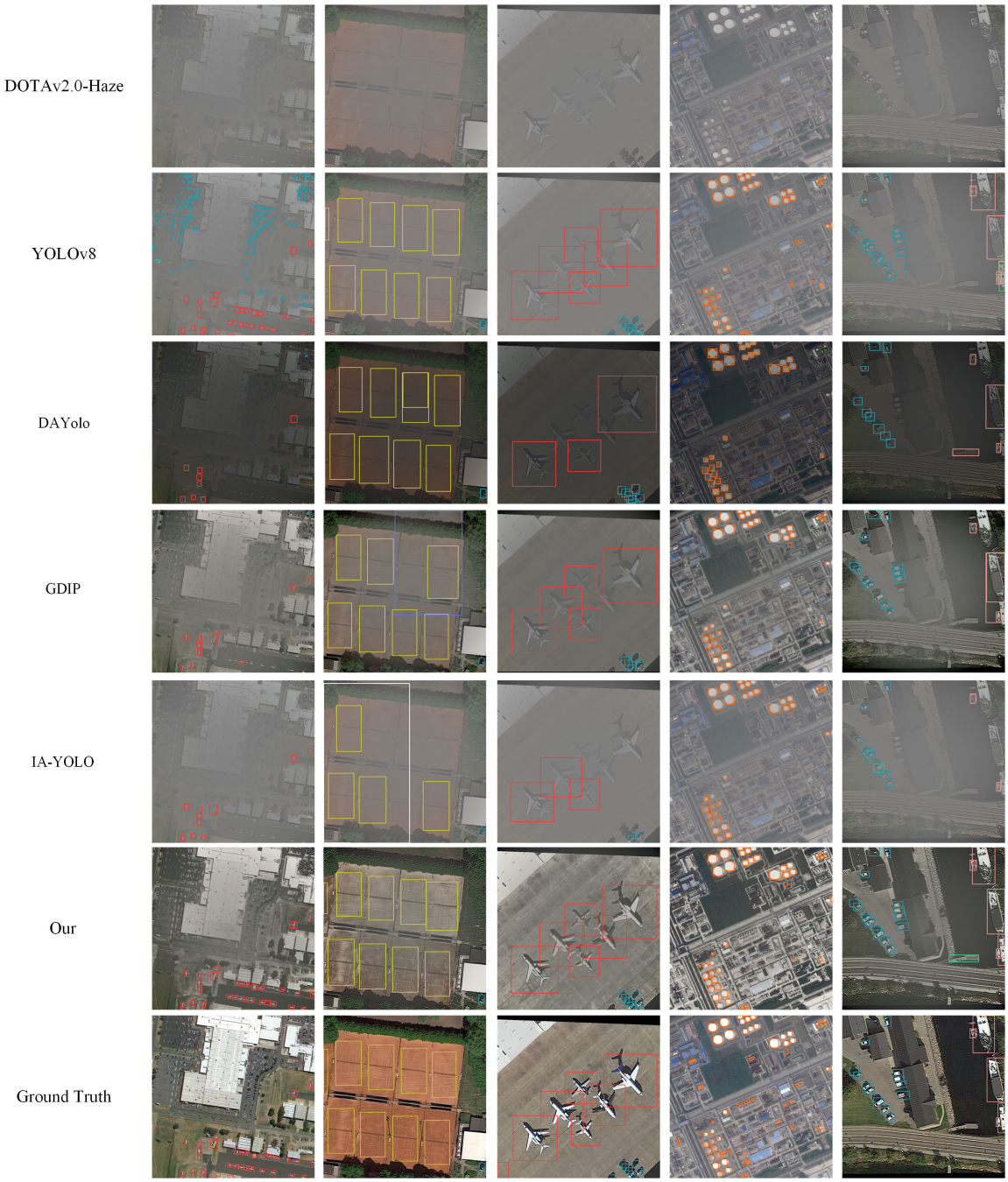


Fig. 6. Comparison of test results visualization.

method in this article performs the best on precision, recall, and mAP50 metrics, reaching 72.8%, 56.0%, and 62.0%, respectively. The baseline method also performs relatively well on precision and recall, with 70.8% and 55.3%, respectively. The GDIP method has a moderate overall performance, while the IA-YOLO and DAYolo methods perform relatively poorly on each indicator. Fig. 6 shows the visualization of the results of the proposed model in comparison with other state-of-the-art algorithms on the DOTA-v2.0Haze dataset, and it can be seen that the model has a better dehazing effect as well as detection performance for remotely sensed haze images in complex climates.

2) Experimental Result on the RTTS Dataset: To verify the generality of this article's method for the hazing case in complex climates, the method was trained on a synthetic dataset containing only person, car, bus, bicycle, and motorcycle in the combination of VOC2007 and VOC2012 (in which the ratio of haze images to clean images is 2:1), with the training epoch set to 80, and the method was tested using the VOCNormal test set (V_N_Ts), synthetic VOChaze test set, and real haze dataset RTTS for testing. YOLOv8 is used as a baseline in the experiments to verify whether the proposed dehaze module is effective or not, and the experimental results are compared with different dehaze

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TABLE V
COMPARISON OF PERFORMANCE BETWEEN NORMAL SYNTHETIC IMAGES AND REAL HAZE IMAGES

Methods		Train Data	V_N_Ts(mAP)	V_F_Ts(mAP)	RTTS(mAP)
Baseline	YOLOv8 [53]	Hybrid	78.69	78.09	49.10
Defog	MSBDN [36]	VOC_Norm	-	57.38	30.30
	GridDehaze [37]	VOC_Norm	-	58.23	31.42
Domain Adaptation	DAYolo [28]	Hybrid	56.51	55.11	29.93
Multi-task	DSNet [38]	Hybrid	53.29	67.40	28.91
Image Adaptive	IA-YOLO [15]	Hybrid	73.23	72.03	37. 08
GDIP [16]	GDIP-YOLO	Hybrid	73.70	71.92	42. 42
	MGDIP-Yolo	Hybrid	75.36	73.77	42.84
	GDIP Regularizer	Hybrid	73.17	72.77	39.52
Proposed Method	DHCNet	Hybrid	81.5	80.9	53.4

TABLE VI
COMPARISON OF DETECTION SPEEDS

Method	FPS	Time Cost(ms)
YOLOv8 [53]	311.40±0.06	3.22±0.00
IA-YOLO [15]	22.83±0.00	43.74±0.00
GDIP-YOLO [16]	29.78±0.10	33.59±0.34
MGDIP-Yolo (top-down) [16]	11.38±0.02	87.88±0.18
GDIP as regularizer [16]	68.39 ± 1.50	14.64±0.69
Our	299.31±0.08	299.31±0.08

object detection and other methods, such as MSBDN [36], GridDehaze [37], DAYOLO [28], DSNet [38], IA-YOLO [15], GDIP [16], and so on, and the experimental results are shown in Table V.

Based on the data in Table V, the method proposed in this article achieves a mean average precision (mAP) of 81.5 in the V_N_Ts dataset, 80.9 in V_F_Ts, and 53.4 in RTTS, outperforming other methods in the table and demonstrating its versatility in detecting objects in both real haze environments and ordinary images. Fig. 7 presents histograms illustrating the performance of different methods across the three datasets, while Fig. 8 compares the test results for VOC and RTTS data using line graphs. Further analysis of the detection performance across various models, as shown in Table V, reveals that the DHCNet significantly outperforms other methods with scores of 81.50–80.90 on V_N_Ts and V_F_Ts, respectively. Overall, DHCNet delivers the best performance across all three metrics, with YOLOv8 and MGDIP-Yolo also performing well. The trend analysis in Fig. 8 offers an intuitive view of relative performance, further confirming the DHCNet as the top performer for the given task.

The detection speed comparison results are shown in Table VI. YOLOv8 has the best performance in terms of the frame rate (FPS), reaching 311.40 FPS, followed by the method proposed in this article at 299.31 FPS, both leading in speed. GDIP-YOLO and IA-YOLO achieve 29.78 and 22.83 FPS, respectively, which are lower than YOLOv8 and our method but still relatively acceptable. The two implementations of MGDIP-YOLO (top-down and GDIP as regularizer) have 11.38 and 68.39 FPS, respectively. Based on the time cost data in the table, GDIP-YOLO and IA-YOLO have time costs of 33.59 and 43.74 ms, respectively, while the two implementations of MGDIP-YOLO (top-down and GDIP as a regularizer) have time costs of 87.88 and 14.64 ms. YOLOv8 and the method proposed in this article exhibit time costs of 3.22 and 3.34 ms, respectively, demonstrating lower per-frame processing times and making them more suitable for tasks requiring high real-time performance. Therefore, YOLOv8 and our method are more suitable for tasks requiring real-time performance, while the other methods may be more appropriate for scenarios with less emphasis on speed but potentially offer advantages in other metrics.

3) Experimental Result on the DOTA-v2.0 Dataset: To prove the effectiveness of centroid-guided contrastive learning on class-imbalanced remote sensing image datasets, the mAP on the DOTA-v2.0 dataset was compared with 16 state-of-the-art remote sensing detection methods. The results are shown in Table VII. The methods compared include BFM [39], DFDet [40], Oriented R-CNN [41], OAN [42], RHINO [43], RRF [44], FCOS-O [45], PDACL [46], DG-YOLOT [47], O-Rep [48], DCFL [49], OASL [50], GRA [51], H2RBox-v2 [52], YOLOv8 [53], and MTP [54]. All methods were trained and tested on the DOTA-v2.0 dataset for 50 epochs. Table VII presents comparative experiments on the DOTA-v2.0 dataset, showing that our proposed method achieves the highest performance in terms of mAP50, with a value of 62.00%, significantly surpassing other methods. To visually depict the performance of each method across all categories in terms of mAP50, Fig. 9 illustrates this through bar charts. Analysis of Fig. 9 reveals that our method achieves the best mAP50 performance at 62.00%, notably outperforming others. Following closely, PDACL [46] achieves an mAP50 of 61.21%. Among traditional two-stage methods, BFM [39] stands out with an mAP50 of 58.69%. In specific category performance, YOLOv8 [53] and PDACL [46] achieve classification accuracies of 91.89% and 83.08%, respectively, in the PL (Plane) category. Our method achieves an accuracy of 84.10% in the PL category, demonstrating similarly excellent performance. These results indicate significant advantages of our method in overall detection performance and specific categories.

4) Experimental Result on the HazeNet Dataset: This section confirms the effectiveness of DHC-Net on UAV data in hazy conditions using the HazeNet dataset [67]. The model was trained for 12 epochs on the training set and tested on both the synthetic test set and the real-world hazy dataset RDDTS. The results in Table VIII show that the proposed method performs best for UAV tasks in hazy environments.

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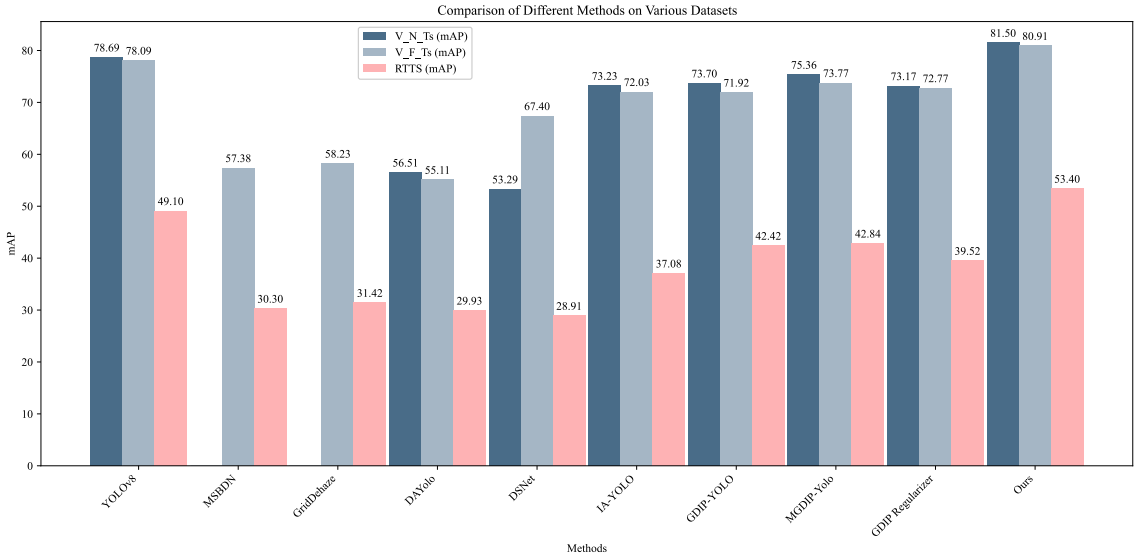


Fig. 7. Plot of comparison results of different methods on different datasets.

TABLE VII

PERFORMANCE EVALUATION OF THE DETECTION TASK ON THE DOTA-V2.0 DATASET. (BOLD INDICATES BEST PERFORMANCE IN EACH COLUMN)

Method	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	CC	AP	HL	mAP50
BFM [39]	80.12	54.12	50.07	65.68	43.98	60.07	67.85	79.11	64.38	60.56	45.98	58.26	58.31	64.82	**69.84**	32.78	**89.37**	11.07	58.69
DFDet [40]	75.44	52.17	42.28	60.17	48.80	53.36	62.67	78.15	56.85	66.62	40.78	53.05	48.42	59.23	51.40	25.47	66.29	16.38	53.19
Oriente R-CNN [41]	78.65	51.80	47.15	65.78	43.25	58.29	60.89	82.83	63.51	59.50	43.40	55.57	52.90	56.18	54.13	27.55	66.24	5.22	54.06
OAN [42]	79.18	51.57	47.50	**66.61**	43.30	58.07	60.73	82.85	64.47	59.62	44.31	56.66	52.71	56.73	53.04	26.01	66.41	14.42	54.64
DG-YOLOT [47]	86.20	70.80	49.90	57.81	60.00	76.90	79.20	93.30	58.40	70.90	79.20	**93.30**	58.40	**70.90**	42.50	**61.30**	67.50	**64.80**	57.91
RHINO [43]	79.32	58.33	46.06	65.79	57.25	58.38	68.53	80.46	63.27	73.01	48.69	57.80	56.21	60.04	59.06	42.70	78.96	12.84	59.26
YOLOv8 [53]	**91.89**	**72.87**	43.42	60.95	62.13	79.11	85.79	94.51	60.78	75.22	42.36	56.53	75.99	68.35	48.51	2.24	33.82	0.15	58.51
RRF [44]	77.58	49.96	38.60	53.82	54.97	57.12	68.93	77.88	59.59	71.92	40.17	51.22	53.00	57.12	49.66	25.42	66.92	5.17	53.28
FCOS-O [45]	74.84	47.53	40.83	57.41	43.89	47.72	55.66	78.61	57.86	63.00	38.02	52.38	41.91	53.24	40.22	7.15	65.51	7.42	48.51
PDACL [46]	83.08	68.53	44.31	58.33	**63. 04**	79.12	88.18	93.87	58.51	72.95	**54.01**	54.84	73.21	57.80	40.70	3.05	61.41	49.67	61.21
O-Rep [48]	73.02	46.68	42.37	63.05	47.06	50.28	58.64	78.84	57.12	66.77	35.21	50.76	48.77	51.62	34.23	6.17	64.66	5.87	48.95
DCFL [49]	79.49	55.97	**50.15**	61.59	49.01	55.33	59.31	81.81	**66.52**	60.06	52.87	56.71	57.83	58.13	6035	35.66	78.65	13.03	57.66
OASL [50]	76.65	55.46	46.33	62.49	53.18	56.62	66.16	80.75	63.07	67.03	44.89	55.68	54.24	59.04	59.15	35.00	77.81	14.65	57.18
H2RBox-v2 [52]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	50.00
GRA [51]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	57.95
MTP [54]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	58.96
Our	84.10	65.40	45.10	61.00	62.90	**79.20**	**88.50**	**95.04**	63.40	**76.00**	51.60	57.50	**76.20**	54.10	31.80	9.87	67.50	49.50	**62.00**

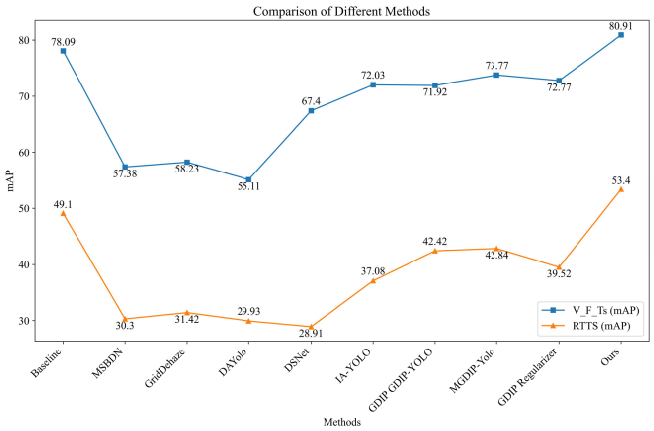


Fig. 8. Comparison of VOC and RTTS data test results.

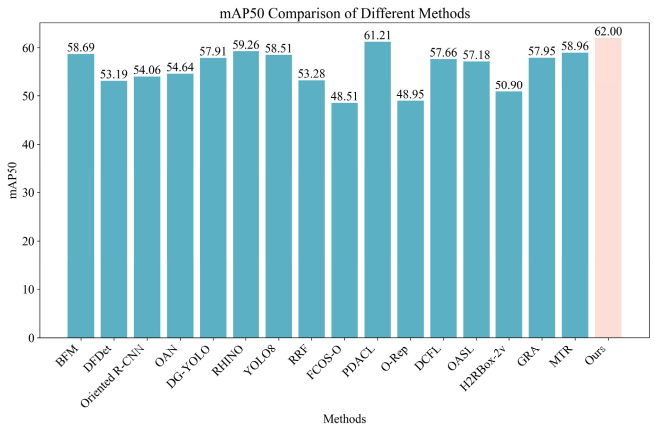


Fig. 9. Comparison of mAP50 results by different methods.

It achieved mAP scores of 67.1% on the test set and 36.1% on RDDTS, surpassing other methods. This highlights the superior detection performance of the model on UAV datasets under hazy conditions.

C. Ablation Studies

1) Investigation of Designed Components: In this section, ablation studies and comprehensive discussions are conducted to demonstrate the effectiveness of the adaptive dehazing module and contrastive learning guided by centroids (CCL).

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TABLE VIII
EXPERIMENTAL RESULT ON THE HAZENET DATASET

Mathod	AP on Test-set				AP on RDDTS			
	Car	Truck	Bus	mAP	Car	Truck	Bus	mAP
Faster RCNN + GridDehaze [37]	48.7	21.2	46.8	38.9	37.6	8.0	13.3	19.6
Faster RCNN + MixDehazeNet [55]	49.5	22.0	48.4	39.9	40.8	8.3	14.4	21.2
Faster RCNN + DSANet [56]	50.1	23.0	49.2	40.8	41.8	9.7	15.8	22.4
Faster RCNN + FFA [13]	50.2	23.5	49.9	41.2	41.1	9.4	15.5	22.0
Faster RCNN + DehazeFormer [14]	51.3	24.7	51.5	42.5	41.2	9.3	15.2	21.9
Faster RCNN + gUNet [57]	51.4	25.3	51.3	42.7	41.7	9.0	15.8	22.2
Faster RCNN + C2PNet [58]	51.5	25.4	51.7	42.9	41.8	9.5	16.0	22.4
Faster RCNN + RIDCP [68]	52.9	26.1	55.4	44.8	43.9	9.7	19.0	24.2
IA-YOLO [15]	44.1	22.2	48.6	38.3	41.9	8.0	17.3	22.4
GDIP [16]	47.0	22.7	50.6	40.1	46.6	8.5	17.8	24.3
TogetherNet [69]	53.4	25.4	55.0	44.6	48.2	11.3	16.1	25.2
FCOS-DeCoDet [67]	55.9	28.6	57.6	47.4	48.1	11.1	17.8	24.3
VFNet-DeCoDet [67]	58.3	33.7	62.5	51.5	49.0	9.0	19.7	25.9
YOLOv8 [53]	85.8	36.6	75.6	65.9	71.9	7.6	22.7	34.1
DHC-Net	**86.6**	**37.1**	**77.5**	**67.1**	**74.9**	**10.4**	**23.0**	**36.1**

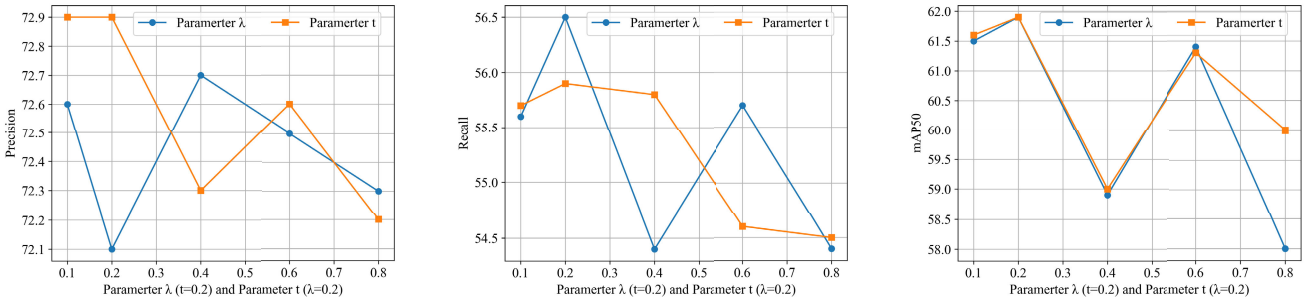


Fig. 10. Key metric results across different parameter settings. (a) Precision. (b) Recall. (c) mAP50.

The aim is to determine the optimal configuration of the proposed method. All experiments are conducted on the DOTA-v2.0Haze test set, with a consistent epoch setting of 50. As shown in Table IX, in the baseline model without any modules, the precision is 70.8%, recall is 55.3%, mAP50 is 57.3%, and mAP95 is 37.2%. After introducing the adaptive dehazing module, all metrics improve, with precision increasing to 72.45% and mAP50 to 58.9%. When the CCL module is added, precision further rises to 72.7% and mAP50 reaches 59.0%. The most significant performance improvement is observed when both the adaptive module and the CCL module are integrated, with precision reaching 72.8%, recall increasing to 56.0%, and mAP50 and mAP95 achieving 62.0% and 40.2%, respectively. These results demonstrate that the combination of the adaptive dehazing module and the CCL module can significantly enhance the overall performance of the detection model. Specifically, the proposed adaptive dehazing module and contrastive learning guided by centroids both contribute to steady performance improvements. Introducing contrastive learning guided by centroids is particularly advantageous for identifying instances with lower counts. The integration of the adaptive dehazing module and contrastive learning guided by centroids achieves optimal results because the adaptive dehazing module generates sufficiently high-quality images, thereby providing higher-quality feature maps and more accurate instance quality indicators.

TABLE IX
DESIGN OF COMPONENT ABLATION EXPERIMENTS

Hazing	Block Dehazing	CCL	Precision(%)	Recall(%)	mAP50(%)	mAP95(%)
			71.80	55.10	58.51	38.21
	✓		71.70	54.10	58.49	38.01
		✓	72.60	55.40	61.80	39.40
	✓	✓	72.66	55.52	61.90	39.43
✓			70.80	55.30	57.30	37.20
✓	✓		72.45	55.60	58.90	38.50
✓		✓	72.70	55.50	59.00	38.60
✓	✓	✓	72.80	56.00	62.00	40.20

2) Weights of Centroid-Guided Contrastive Learning Loss: This section analyzes the impact of the loss of weight for centroid-guided contrastive learning on the overall performance of the model through a series of experiments. The experimental results are shown in Table X. From the results, it is evident that too little or too much emphasis on the global contrastive loss negatively affects the model's performance. Therefore, in this experiment, the loss weight is set to 0.2 to

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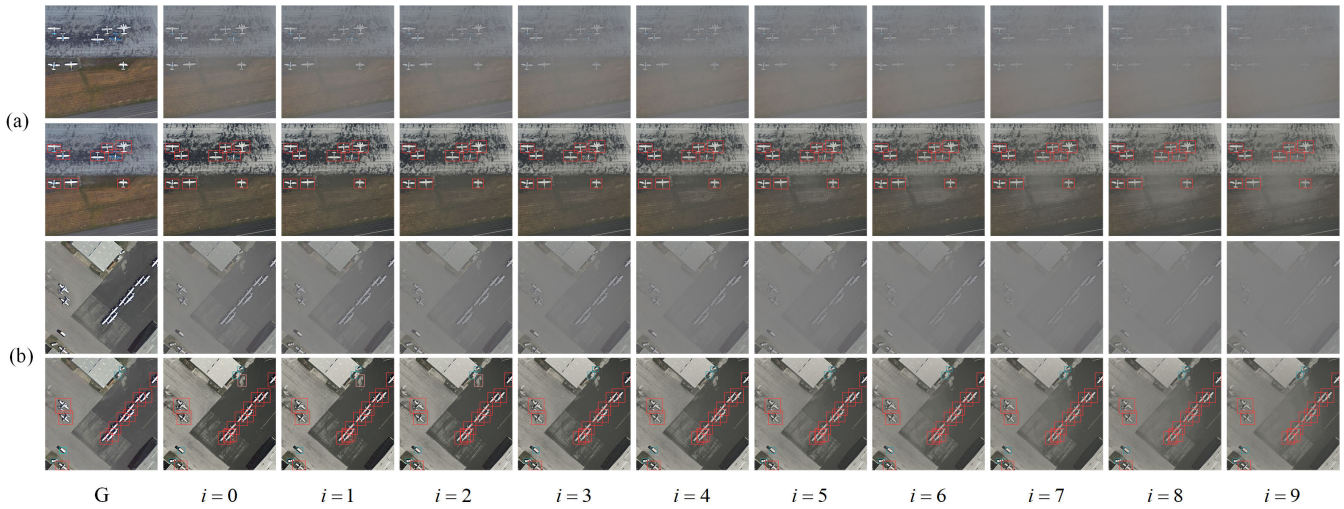


Fig. 11. Detection results of the DHC-Net under different haze concentrations (0–9 represent different concentrations of haze). (a) and (b) Represents the comparison of defogging conditions of two images with different backgrounds and different haze concentrations.

TABLE X
IMPACT OF CONTRASTIVE LEARNING LOSS WEIGHTS ON MODEL PERFORMANCE

λ	Precision(%)	Recall(%)	mAP50(%)	mAP95(%)
0.1	72.6	55.6	61.5	39.0
0.2	72.1	56.5	61.9	39.7
0.4	72.7	54.4	58.9	38.0
0.6	72.5	55.7	61.4	39.9
0.8	72.3	54.4	58.0	37.2

TABLE XI
IMPACT OF THE TEMPERATURE PARAMETER ON MODEL PERFORMANCE

τ	Precision(%)	Recall(%)	mAP50(%)	mAP95(%)
0.1	72.9	55.7	61.6	39.3
0.2	72.9	55.9	61.9	40.1
0.4	72.3	54.8	59.0	37.6
0.6	72.6	55.6	61.3	39.9
0.8	72.2	54.5	60.0	38.4

TABLE XII
PERFORMANCE OF THE PROPOSED METHOD ON THE DOTA-V2.0HAZE DATASET

Category	Precision(%)	Recall(%)	mAP50(%)	mAP95(%)
Plane	92.8	75.9	84.1	61.2
Ship	92.5	82.5	88.9	62.0
Storage tank	87.7	61.5	75.0	48.4
Baseball diamond	72.2	65.1	65.4	37.6
Tennis court	90.6	91.1	94.2	85.1
Basketball court	76.1	59.7	63.4	47.5
Ground track field	70.8	56.1	61.0	41.2
Harbor	82.1	67.8	75.2	43.3
Bridge	63.9	43.2	45.1	21.1
Large vehicle	79.5	72.5	79.0	58.0
Small vehicle	77.4	55.3	62.9	33.1
helicopter	23.6	46.8	31.8	16.2
Roundabout	77.8	49.2	57.5	37.2
Soccer ball field	61.6	52.0	51.6	36.8
Swimming pool	74.8	46.7	54.1	23.8
Container crane	21.3	13.3	9.87	5.2
airport	66.7	68.2	67.5	39.1
Helipad	1.0	0.0	49.5	27.3
all	72.8	56.0	62.0	40.2

ensure optimal overall accuracy. The performance of the detection model varies with different values of λ. The results indicate that precision reaches its highest value of 72.7% when λ is 0.4, while recall reaches its highest value of 56.5% when iii is 0.2. Additionally, mAP50 achieves its highest value of 61.9% at λ equal to 0.2, and mAP95 reaches its highest value of 39.9% at iii equal to 0.6. These findings suggest that λ values of 0.2 and 0.6 exhibit superior overall performance across various metrics, effectively enhancing the overall performance of the detection model. The setting of the temperature parameter τ in contrastive learning is crucial. To determine the optimal choice of τ, a series of experiments were designed, as shown in Table XI. The results indicate that when τ ranges from 0.1 to 0.8, the overall mAP initially increases to 61.9% and then decreases to 60.0%. Therefore, the centroid-guided contrastive learning proposed in this article selects τ as 0.2. Fig. 10 depicts the precision, recall, and mAP evaluations under different parameter settings, revealing a nonmonotonic trend. Based on the results in Fig. 10, the final optimal parameters selected for the model are $\lambda = 0.2$ and $\tau = 0.2$.

D. Analysis of Results

This article analyzes the experimental results on the DOTA-v2.0Haze dataset, with detailed results shown in Table XII. Overall, there are significant differences in detection performance across different categories. The tennis court category performs the best across all metrics, with the highest recall rate of 91.1% and mAP50 of 94.2%, which can be attributed to its distinct rectangular shape and consistent color patterns even under haze interference. The plane and ship categories excel in precision, at 92.8% and 92.5%, respectively. However, the helipad and container cranes categories perform the worst, with helipad having a recall rate of 0% and container cranes having an mAP50 of only 9.87%, with their inferior detection performance primarily arising from their nature as small-sized targets, along with substantial intraclass variation in object

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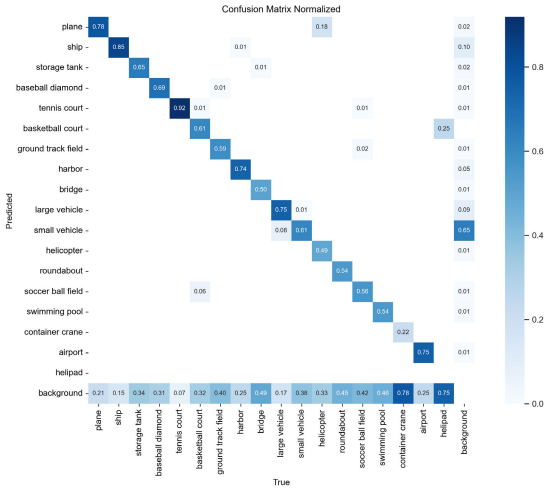


Fig. 12. Test results of the DHC-Net dataset.

shapes and viewing angles. The overall averages show a global precision of 72.8%, recall of 56.0%, mAP50 of 62.0%, and mAP95 of 40.2%. The DHC-Net was thoroughly tested on the synthetic haze DOTA-v2.0Haze dataset, using images with different haze levels, as shown in Fig. 11. The results show that the DHC-Net effectively removes haze from these images and achieves high accuracy in object detection. This demonstrates that the DHC-Net excels in processing remote sensing images under challenging weather conditions. It not only clears the haze effectively but also maintains high detection performance, offering reliable technical support for real-world applications.

The normalized confusion matrix further validates these findings, as shown in Fig. 12. For the helipad category, a significant proportion of misdetections 83% were mistakenly identified as circular farmland due to shape similarities at smaller scales. Meanwhile, container cranes were predominantly misclassified as gantry cranes, with an error rate of 68%. This underscores the need for fine-grained feature disentanglement. Overall, the model demonstrates strong performance across most categories, especially for classes like airplanes, ships, and tennis courts. However, while the performance is generally well-balanced, improvements are still needed for more complex categories such as helipad and container cranes. Future research will focus on exploring few-shot learning approaches to further address the data imbalance challenges.

VI. Conclusion

This article introduces the DHC-Net, a network designed specifically for object detection in remote sensing images under challenging weather conditions, tackling the often-overlooked issues of haze interference and class imbalance in existing algorithms. By incorporating an adaptive dehazing preprocessing module, the DHC-Net enhances image quality and boosts detection performance in hazy conditions. The centroid-guided contrastive learning strategy further reduces the misclassification of minor class samples, ensuring accurate detection of underrepresented categories. The experimental results show that the proposed method outperforms others in scenarios involving haze and imbalanced remote sensing images. In future research, the focus will broaden to encompass other complex climatic conditions, such as rain, snow, and fog. These advancements will help develop more resilient and widely applicable remote sensing image detection systems, ultimately enhancing data acquisition and analysis in various fields, including agriculture, urban planning, and disaster management.

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